Derek Wang

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Technical Skills

Languages: C++, Java, Python, JavaScript, TypeScript, SQL

Frameworks: React, Next.js, Node.js, Express, FastAPI

Libraries/Tools: PyTorch, OpenCV, SQLite, PostgreSQL, MongoDB, Docker, Git, ROS2, Foxglove, Fusion 360

Certifications: Docker Foundations Professional

Experience

Autonomous Self Driving Software Engineer — Sept 2025 – Present

WATonomous — *Waterloo, ON, Canada*

- Architected and deployed ROS2 navigation stack (costmaps, A* planning, Pure Pursuit) enabling autonomous rover path following in dynamic environments (hills, obstacles, holes), simulating Mars environment
- Led YOLO11s model retraining, improving detection mAP from 30% to 80% across 3,448 labeled images via hyperparameter tuning and augmentation redesign
- Optimized YOLO11s inference to 30 FPS on rover hardware via ONNX export and onnxslim optimization

Autonomy Developer — Aug 2025 – Feb 2026

Waterloo Aerial Robotics Group — *Waterloo, ON, Canada*

- Developed Python hardware-in-the-loop UAV simulation using Ardupilot Mission Planner, Pixhawk flight controller and MAVLink protocol for the 2026 UAS Student Competition
- Eliminated GPS teleportation bugs upon injection and achieved smooth trajectory interpolation at 24 FPS via periodic position updates, drone speed control, position thresholding, and trajectory smoothing algorithms

Robotics Software Programmer – VEX Robotics Competition — Sept 2024 – June 2025

Checkmate Robotics Club — *Markham, Ontario*

- Reduced autonomous robot positioning error to ±0.5cm by implementing wheel odometry, distance sensors, and IMU for position estimation, and PID control with trapezoidal motion profiling algorithms for trajectory control
- Qualified for 2025 VEX World Championships (top 10% globally) and placed 5th/80 at Ontario Provincials

Projects

Reparo - AI Repair Tool (Hack Canada 2026 Winner) | *Python, FastAPI, Swift, Gemini, SerpAPI, Reactiv*

- Generated 5+ structured repair outputs from a single device image using a computer-vision + Gemini reasoning pipeline that performs component detection and contextual repair inference for consumer devices
- Built a FastAPI orchestration service that normalizes unstructured LLM responses into a typed repair schema, enabling deterministic UI rendering of repair steps, cost estimates, and required tools
- Designed a multi-stage parts retrieval pipeline (SerpAPI → Shopify storefront discovery → Shopify product APIs) to automatically source purchasable repair components surfaced in a Swift Reactiv App Clip

Silhouette - Cursor for Fashion | *Node.js, Next.js, Express, MongoDB, Gemini, Overshoot.ai SDK*

- Delivered personalized fashion recommendations in under 2s at by building Gemini pipelines with MERN stack that orchestrate async inference, rate limits, product recommendations, and style constraints over live camera input
- Secured a VC referral to continue development as a Proof of Concept (Tailor AI) for Merian Brothers

Hierarchical Reasoning Retrieval | *Node.js, Express, SQLite, Gemini*

- Designed hierarchical LLM document retrieval system traversing heading trees instead of flat vector similarity
- Decreased LLM token usage by up to 80% and optimized retrieval by engineering an SQLite pipeline that chunks markdown into document trees using bottom-up summarization, keyword pre-filtering, and tree search
- Implemented beam search traversal over document heading trees for structured document search

Autonomous Vehicle Control System | *C++, ROS2, Docker, CMake, Foxglove*

- Built a C++ LiDAR perception pipeline generating real-time occupancy costmaps at 0.1m resolution in Foxglove
- Developed ROS2 autonomous navigation stack with Docker, validated 100% obstacle avoidance in 50+ simulations

Education

University of Waterloo — Expected Apr 2030

Bachelor of Applied Science in Computer Engineering (Co-op) — *GPA: 3.7*